

Asymptotic statistics on stratified spaces

Lecture 1: Central limit theory and Fréchet means

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Welcome

Welcome to “Asymptotic statistics on stratified spaces”!

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This is a new course; I welcome your feedback and questions.

Statistics: what is it?

Statistics is the *scientific* and *mathematical* field that studies how to connect empirical observations to broader phenomena – scientific, medical, economic, athletic. . .

Statistics: a typical setup

- ▶ We have n observations $X_1, X_2, \dots$ which live in a sample space $\mathcal{X}$.
- ▶ We assume that observations are drawn independently from some distribution $P \in \mathcal{P}$, which we don't know.
- ▶ We want to learn about some summary of P , $\theta = \theta(P) \in \Theta$.
 - ▶ $\theta : \mathcal{P} \rightarrow \Theta$
- ▶ To do this, we construct a function $\hat{\theta}_n$
- ▶ Our best estimate of $\theta(P)$ is $\hat{\theta}_n(P_n)$, where P_n is the distribution of $X_1, X_2, \dots, X_n$

What can we promise about $\hat{\theta}_n(P_n)$? How well does it reflect $\theta(P)$?

Asymptotic statistics: what is it?

Without making strong assumptions on P (e.g., fully constraining the shape of P , leaving only a small number of parameters unknown), it's hard to talk about the behavior of $\hat{\theta}_n(P_n)$ for specific, finite n .

Instead, we study the asymptotic behavior of $\hat{\theta}_n(P_n)$ as $n \rightarrow \infty$.

Asymptotic statistics: what is it?

Overarching questions:

- Consistency: Is $\hat{\theta}_n(P_n)$ guaranteed to get closer and closer to $\theta(P)$ as $n \rightarrow \infty$?
- Variation: Does the variation in $\hat{\theta}_n(P_n)$ get smaller and smaller? At what rate?
- Distribution: Can we know/approximate the probability that $\hat{\theta}_n(P_n)$ is distance ϵ away from $\theta(P)$?

Why study asymptotic statistics?

These questions can guide choices of $\hat{\theta}_n$. For example,

- ▶ Is it a good idea to set $\hat{\theta}_n = \theta$?
 - ▶ When and why?
 - ▶ If not, how can we construct a better $\hat{\theta}_n$?

Optimal estimators allow us to take maximum advantage of the observations that we have.

Asymptotic statistics and optimality

Optimality of estimators is well-understood when $\mathcal{X} = \mathbb{R}^d$ and $\Theta = \mathbb{R}^p$. In general,

- ▶ Consistent estimators can usually be found
- ▶ Under restrictions on P , we can find a $\hat{\theta}_n$ with minimum variance out of all consistent estimators (*efficient*).
 - ▶ In general, the greater the restrictions on P , the smaller this variance is.
- ▶ If we assume that P is in some class *but we are wrong*, we can study the behavior of $\hat{\theta}_n$ with respect to both its limit and its distribution.

Open questions in asymptotic statistics

Optimality of estimators for more general $\mathcal{X}$ and Θ is *not well-understood*.

Thankfully, many of the needed mathematical tools has been developed.

We will review some key results in this minicourse.

But first, some motivation from applied statistics. . .

Applied motivation: assessing strength of evidence

Asymptotic statistics (here, central limit theory) is used to assess the strength of evidence in *every single modern study of therapeutics*, because it allows us to answer the question:

How unlikely is what we saw if the treatment didn't help?

Applied motivation: assessing strength of evidence

In a randomized trial of a new treatment for blood-pressure (reduction in mmHg after 8 weeks),

- ▶ 20 people received treatment: $\bar{X}_1 = 7.0$, $s_1 = 3.0$
- ▶ 20 people received placebo: $\bar{X}_0 = 4.0$, $s_0 = 3.0$

If the average effect were the same, then by results we will review shortly (CLT + Slutsky),

$$Z := \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}} \sim N(0, 1).$$

Applied motivation: assessing strength of evidence (ctd)

Here $Z = 3.0 / \sqrt{9/20 + 9/20} \approx 3.16$, so seeing a result *as or more surprising* has probability

$$\Pr(|Z| \geq 3.16) = 2(1 - \Phi(3.16)) \approx 0.0016.$$

That's very surprising! It makes me strongly question the premise that treatment and placebo have the same average reduction in blood pressure.

“Reject the hypothesis.”

Applied motivation: assessing strength of evidence (ctd)

No assumption on the distribution of the observations was needed!

We only needed the asymptotic approximation

$$\frac{\bar{X}_1 - \bar{X}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}} \stackrel{\sim}{\sim} N(0, 1)$$

to be close.

My point: *We can do use tools from asymptotic statistics to answer the most important questions in applied statistics.* E.g., to approximate the probability that $\hat{\theta}_n(P_n) = \bar{X}_1 - \bar{X}_0$ is distance ϵ away from $\theta(P) = \mu_1 - \mu_0$!

Applied motivation: model misspecification

Here two other examples of where asymptotic statistics gives us insight into important applied questions:

1. Model misspecification

- ▶ Logistic regression is often used to study how the probability of an event differs between groups
- ▶ $\Pr(\text{event}) = \frac{\exp(X_i^T \beta)}{1 + \exp(X_i^T \beta)}$, equivalently, $\log(\text{odds}) = X_i^T \beta$.
- ▶ If the log-odds of the event is not actually linear in X_1 , does $\hat{\beta}_1$ have any meaning?

Applied motivation: efficiency

2. Efficiency

- ▶ We would like to estimate the average treatment effect under assignment: $\mathbb{E}[Y(1) - Y(0)]$
- ▶ Who self-selects treatment may differ in their underlying characteristics X , and $X \rightarrow Y$
- ▶ Approach:
 - ▶ estimate $Q(a, x) = \mathbb{E}[Y|X = x, \text{chose tmt } a]$
 - ▶ integrate Q wrt a target covariate distribution:
$$\int Q(1, x) - Q(0, x) dP(x)$$
 - ▶ “standard causal assumptions” (eg unmeasured confounding) give us that this targets the ATE

Applied motivation: efficiency

2. Efficiency (ctd)

- ▶ If we estimate Q inflexibly, our estimate will be inconsistent (meaningless?)
- ▶ If we estimate Q flexibly, our estimate will converge more slowly than $n^{-1/2}$
- ▶ A functional Taylor expansion gives us a new, corrected estimate that
 - ▶ addresses the first-order bias
 - ▶ obtains the “parametric” rate of $n^{-1/2}$
 - ▶ has *the smallest possible variance for a consistent estimator* for our estimate of $\mathbb{E}[Y(1) - Y(0)]$

My motivation

Open questions:

Which of these ideas from $\mathbb{R}^d$ can apply to more flexible sample and parameter spaces?

Which are useful in practice?

Key definitions: Modes of convergence

Let Z_n, Z be random elements of a metric space (S, ρ) .

- ▶ $Z_n \xrightarrow{\mathcal{D}} Z$ if $\mathbb{E}f(Z_n) \rightarrow \mathbb{E}f(Z)$ for every bounded continuous $f : S \rightarrow \mathbb{R}$.
- ▶ $Z_n \xrightarrow{P} Z$ if $\lim_{n \rightarrow \infty} \Pr(\rho(Z_n, Z) > \varepsilon) = 0$ for every $\varepsilon > 0$.
- ▶ $Z_n \xrightarrow{a.s.} Z$ if $\Pr\left(\lim_{n \rightarrow \infty} \rho(Z_n, Z) = 0\right) = 1$.

Useful shorthand:

- ▶ $Z_n = O_p(a_n)$ if $\lim_{K \rightarrow \infty} \limsup_n \Pr(\|Z_n\| > Ka_n) = 0$
 - ▶ $Z_n = O_p(1)$ if $\lim_{K \rightarrow \infty} \limsup_n \Pr(\|Z_n\| > K) = 0$
- ▶ $Z_n = o_p(a_n)$ if $\|Z_n\|/a_n \xrightarrow{P} 0$.
 - ▶ $Z_n = o_p(1)$ if $Z_n \xrightarrow{P} 0$.

Key tools: Convergence in distribution and probability

Lemma: (Slutsky) If $Z_n \xrightarrow{\mathcal{D}} Z$ in $\mathbb{R}^d$ and $A_n \xrightarrow{P} A$ (constant matrix), $B_n \xrightarrow{P} b$ (constant vector), then $A_n Z_n + B_n \xrightarrow{\mathcal{D}} AZ + b$.

Lemma: (Continuous mapping) If $Z_n \xrightarrow{\mathcal{D}} Z$ and g is continuous a.e.- P_Z , then $g(Z_n) \xrightarrow{\mathcal{D}} g(Z)$.

Theorem: (Law of large numbers, LLN) If $X_1, X_2, \dots$ are i.i.d. in $\mathbb{R}^d$ with $\mathbb{E}|X_1| < \infty$, then $\bar{X}_n \xrightarrow{a.s.} \mathbb{E}X_1$ (and hence $\bar{X}_n \xrightarrow{P} \mathbb{E}X_1$).

Useful notation: Empirical measures

Empirical measure notation: for $f : \mathcal{X} \rightarrow \mathbb{R}^k$,

$$P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad Pf = \int f \, dP,$$

Lemma: (Dominated convergence) If $f_n(z) \rightarrow f(z)$ pointwise P_Z -a.e. and $|f_n| \leq g$ a.e.- P_Z with $Pg < \infty$, then $Pf_n \rightarrow Pf$.

Central Limit Theorem in $\mathbb{R}^d$: Statement

“The”* *central limit theorem*, one of the most important results in statistics.

Theorem: Let $X_1, X_2, \dots$ be i.i.d. random vectors in $\mathbb{R}^d$ with $\mathbb{E}\|X_1\|^2 < \infty$, $\mu = \mathbb{E}X_1$ and $\Sigma = \text{Var}(X_1) = \mathbb{E}(X_1 - \mu)(X_1 - \mu)^T$. Then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N_d(0, \Sigma), \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

* “The” = multivariate Lindeberg–Lévy; there are actually *many* CLTs.

Comments

Before we (sketch) prove it – why is this remarkable

- ▶ centering
- ▶ variance scaling
- ▶ exact shape of tails – all moments known, asymptotically
 - ▶ essential for assessing how “surprising” a result is; a backbone of modern scientific reasoning

Tools: characteristic functions

For a random vector Z in $\mathbb{R}^d$, $\varphi_Z(t) = \mathbb{E}e^{it^T Z}$.

Theorem: (Lévy continuity) If $\varphi_{Z_n}(t) \rightarrow \varphi(t)$ for every $t \in \mathbb{R}^d$ and φ is continuous at 0, then φ is the characteristic function of some Z and $Z_n \xrightarrow{\mathcal{D}} Z$.

Lemma: (Cramér–Wold) $Z_n \xrightarrow{\mathcal{D}} Z$ in $\mathbb{R}^d$ if and only if $t^T Z_n \xrightarrow{\mathcal{D}} t^T Z$ in $\mathbb{R}$ for every $t \in \mathbb{R}^d$.

Proof: : $(\Rightarrow)$ continuous mapping. $(\Leftarrow)$: change of variable + Lévy continuity

Tools: a Taylor expansion

Lemma: For any Y such that $\mathbb{E}Y^2 < \infty$, as $s \rightarrow 0$,

$$\varphi_Y(s) = 1 + is\mathbb{E}Y - \frac{1}{2}s^2\mathbb{E}Y^2 + o(s^2).$$

Proof: Taylor series for e^{iu} , let $u = sY$, take expectations and apply DCT

Central Limit Theorem in $\mathbb{R}^d$: Proof

Fix $t \in \mathbb{R}^d$ and set $Y_i = t^T(X_i - \mu)$, so $\mathbb{E} Y_i = 0$ and $\text{Var}(Y_i) = t^T \Sigma t =: \sigma^2 < \infty$. Then

$$t^T \sqrt{n}(\bar{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i.$$

By independence and the Lemma, for fixed s ,

$$\varphi_{n^{-1/2} \sum Y_i}(s) = \left[\varphi_Y(s/\sqrt{n}) \right]^n = \left[1 - \frac{\sigma^2 s^2}{2n} + o(n^{-1}) \right]^n.$$

Proof (continued)

Using $(1 + c_n/n)^n \rightarrow e^c$,

$$\varphi_{n^{-1/2}} \sum \gamma_i(s) \longrightarrow e^{-\sigma^2 s^2/2},$$

the characteristic function of $N(0, \sigma^2)$, which is continuous at 0.
By Lévy continuity,

$$t^T \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N(0, t^T \Sigma t) \stackrel{d}{=} t^T N_d(0, \Sigma).$$

Cramér–Wold then gives $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N_d(0, \Sigma)$. $\square$

How specific is the CLT?

For $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, we have that $\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N_d(0, \Sigma)$.

Not every target parameter of interest is an *average* of random vectors.

Can we get amazing distributional results for other types of parameters?

A CLT for minimizers?

Setting: $X_1, X_2, \dots \sim P$, and we have a function $m(\theta, X)$ whose minimizer we'd like to know:

$$\theta_0 = \arg \min_{\theta \in \Theta} \int m(\theta, X) dP = \arg \min_{\theta \in \Theta} P m_\theta$$

We estimate θ using the sample minimizer:

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n m(\theta, X_i) = \arg \min_{\theta \in \Theta} P_n m_\theta.$$

How does $\hat{\theta}_n$ behave for large n ?

Notation

$\Theta \subseteq \mathbb{R}^s$ open; $x \mapsto m_\theta(x)$ measurable for each θ ;

$$H(\theta) = Pm_\theta, \quad H_n(\theta) = P_n m_\theta.$$

Write $D_r = \partial/\partial\theta_r$ and $D_{r,r'} = D_r D_{r'}$, and, where these exist,

$$\dot{m}_\theta = (D_1 m_\theta, \dots, D_s m_\theta)^T \in \mathbb{R}^s, \quad \ddot{m}_\theta = [D_{r,r'} m_\theta]_{r,r'} \in \mathbb{R}^{s \times s}.$$

Fix $\theta_0 \in \Theta$ and let $B_\delta = \{\theta \in \mathbb{R}^s : \|\theta - \theta_0\| \leq \delta\}$.

$\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^s$ and the spectral norm on $\mathbb{R}^{s \times s}$.

Conditions

- (M1) **Identification.** $\theta_0 \in \Theta$ is the unique minimiser of H over Θ , and $H(\theta_0) < \infty$.
- (M2) **Smoothness.** For P -a.e. x , $\theta \mapsto m_\theta(x)$ is twice continuously differentiable on Θ .
- (M3) **Localisation.** $\hat{\theta}_n \xrightarrow{P} \theta_0$, and $\Pr(H_n(\hat{\theta}_n) = \inf_{\theta \in \Theta} H_n(\theta)) \rightarrow 1$.
- (M4) **Moments.** $P|D_r m_{\theta_0}|^2 < \infty$ and $P|D_{r,r'} m_{\theta_0}| < \infty$ for $1 \leq r, r' \leq s$.
 Write

$$\Lambda = [PD_{r,r'} m_{\theta_0}]_{r,r'} = P\ddot{m}_{\theta_0}, \quad \Sigma = P\dot{m}_{\theta_0}\dot{m}_{\theta_0}^T.$$

- (M5) **Locally uniform L^1 -smoothness of the Hessian.** With

$$u_{r,r'}(\delta; x) = \sup_{\theta \in B_\delta} |D_{r,r'} m_\theta(x) - D_{r,r'} m_{\theta_0}(x)|, \quad \gamma_{r,r'}(\delta) = P u_{r,r'}(\delta; \cdot),$$

one has $\gamma_{r,r'}(\delta) \rightarrow 0$ as $\delta \downarrow 0$, for all r, r' .

- (M6) **Nondegeneracy.** Λ is nonsingular.

Asymptotic linearity

Definition: An estimator $\hat{\theta}_n$ of θ_0 is *asymptotically linear at P* with influence function $\varphi : \mathcal{X} \rightarrow \mathbb{R}^s$ if

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(X_i) + o_p(1).$$

for φ such that $P\varphi = 0$ and $P\|\varphi\|^2 < \infty$.

Asymptotically linear estimators are, to first order, *averages*.

Note: $\varphi(x)$ is the first-order effect on $\hat{\theta}_n$ of one observation at x : writing $\hat{\theta}_n = T(P_n)$,

$$\varphi(x) = \lim_{t \downarrow 0} \frac{T((1-t)P + t\delta_x) - T(P)}{t}$$

Theorem: a CLT for M-estimators

Theorem: Assume (M1)–(M6). Then $\hat{\theta}_n$ is asymptotically linear with influence function $-\Lambda^{-1}\dot{m}_{\theta_0}$, i.e.,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\Lambda^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{m}_{\theta_0}(X_i) + o_p(1) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1} \Sigma \Lambda^{-1}),$$

where

$$\Sigma = P \dot{m}_{\theta_0} \dot{m}_{\theta_0}^T = \text{Var}(\dot{m}_{\theta_0}(X_1)).$$

Theorem: a CLT for M-estimators

$$\text{WTS: } \sqrt{n}(\hat{\theta}_n - \theta_0) = -\sqrt{n}\Lambda^{-1}P_n\dot{m}_{\theta_0} + o_p(1)$$

Proof overview:

1. $P_n\dot{m}_{\hat{\theta}_n} = 0$ on a set with probability $\rightarrow 1$
2. $P_n\dot{m}_{\hat{\theta}_n} = P_n\dot{m}_{\theta_0} + \Lambda_n h_n$ for $h_n = \hat{\theta}_n - \theta_0$ and some Λ_n
3. $\Lambda_n \xrightarrow{P} \Lambda$
4. $h_n = -\Lambda_n^{-1} P_n\dot{m}_{\theta_0} = -(\Lambda^{-1} + o_p(1))P_n\dot{m}_{\theta_0}$
5. Apply CLT: $\sqrt{n} P_n\dot{m}_{\theta_0} \xrightarrow{\mathcal{D}} N(0, \Sigma)$

We will use without (detailed) proof that $\nabla H(\theta_0) = P\dot{m}_{\theta_0} = 0$.

$$1. P_n \dot{m}_{\hat{\theta}_n} = 0$$

By (M2), $\theta \mapsto m_\theta(x)$ is C^2 on Θ a.e., so $H_n(\theta) = P_n m_\theta$ is C^2 on Θ a.e.

Fix δ_0 and let

$$A_n = \left\{ H_n(\hat{\theta}_n) = \inf_{\theta \in \Theta} H_n(\theta) \right\} \cap \{ \|\hat{\theta}_n - \theta_0\| \leq \delta_0 \}.$$

By (M3), $\Pr(A_n) \rightarrow 1$.

On A_n , $\hat{\theta}_n$ minimises H_n (differentiable) over Θ (open), so $P_n \dot{m}_{\hat{\theta}_n} = 0$.

$$2. P_n \dot{m}_{\hat{\theta}_n} = P_n \dot{m}_{\theta_0} + \Lambda_n h_n$$

Define $h_n := \hat{\theta}_n - \theta_0$.

On A_n , $[\theta_0, \hat{\theta}_n] \in B_{\delta_0}$ (convex) and $t \mapsto P_n \dot{m}_{\theta_0 + th_n}$ is C^1 on $[0, 1]$ with derivative $(P_n \ddot{m}_{\theta_0 + th_n}) h_n$. So the fundamental theorem gives

$$P_n \dot{m}_{\hat{\theta}_n} = P_n \dot{m}_{\theta_0} + \left(\int_0^1 P_n \ddot{m}_{\theta_0 + th_n} dt \right) h_n, =: P_n \dot{m}_{\theta_0} + \Lambda_n h_n$$

3: $\Lambda_n \xrightarrow{P} \Lambda$

Recall $\Lambda := P\ddot{m}_{\theta_0}$. We show $(\Lambda_n)_{r,r'} \xrightarrow{P} (\Lambda)_{r,r'}$ for every r, r' .

$$(\Lambda_n)_{r,r'} - P_n D_{r,r'} m_{\theta_0} = \int_0^1 P_n (D_{r,r'} m_{\theta_0+th_n} - D_{r,r'} m_{\theta_0}) dt.$$

$$\begin{aligned} |(\Lambda_n)_{r,r'} - P_n D_{r,r'} m_{\theta_0}| &\leq \int_0^1 P_n |D_{r,r'} m_{\theta_0+th_n} - D_{r,r'} m_{\theta_0}| dt \\ &\leq P_n u_{r,r'}(\delta; \cdot) \end{aligned}$$

(by our local uniform assumption) on $A_{n,\delta} = A_n \cap \{\|h_n\| \leq \delta\}$ for $\delta \in (0, \delta_0]$.

3 (ctd): $\Lambda_n \xrightarrow{P} \Lambda$

Recall Markov's inequality: For nonnegative Y with $\mathbb{E}Y < \infty$, $\Pr(Y \geq \varepsilon) \leq \frac{\mathbb{E}Y}{\varepsilon}$ for any $\varepsilon > 0$.

$u_{r,r'}(\delta; \cdot)$ is nonnegative and $\mathbb{E}[P_n u_{r,r'}(\delta; \cdot)] = \gamma_{r,r'}(\delta)$, so fixing $\varepsilon > 0$ gives

$$\Pr\left(|(\Lambda_n)_{r,r'} - P_n D_{r,r'} m_{\theta_0}| > \varepsilon\right) \leq \Pr(A_{n,\delta}^c) + \gamma_{r,r'}(\delta)/\varepsilon.$$

Fix δ . By (M3), $\Pr(A_{n,\delta}^c) \rightarrow 0$. Then let $\delta \downarrow 0$. By (M5),

$$(\Lambda_n)_{r,r'} - P_n D_{r,r'} m_{\theta_0} = o_p(1).$$

By (M4) and the LLN, $P_n D_{r,r'} m_{\theta_0} \xrightarrow{P} \Lambda_{r,r'}$. As r, r' were arbitrary,

$$\Lambda_n \xrightarrow{P} \Lambda.$$

4: Invert and apply CLT

The invertible matrices form an open subset of $\mathbb{R}^{s \times s}$ on which $A \mapsto A^{-1}$ is continuous. By (M6) and continuous mapping, $\Lambda_n^{-1} \xrightarrow{P} \Lambda^{-1}$.

Since $0 = P_n \dot{m}_{\theta_0} + \Lambda_n h_n$, we have

$$\sqrt{n} h_n = -\sqrt{n} \Lambda_n^{-1} P_n \dot{m}_{\theta_0} = -\sqrt{n} (\Lambda^{-1} + o_p(1)) P_n \dot{m}_{\theta_0}$$

The CLT gives $\sqrt{n} P_n \dot{m}_{\theta_0} \xrightarrow{\mathcal{D}} N(0, \Sigma)$, so

$$\sqrt{n}(\hat{\theta}_n - \theta) = -\sqrt{n} \Lambda^{-1} P_n \dot{m}_{\theta_0} + o_p(1) O_p(1) = -\sqrt{n} \Lambda^{-1} P_n \dot{m}_{\theta_0} + o_p(1).$$

This is asymptotic linearity, and Slutsky gives the limit law. $\square$

Generalizing averages to $\mathcal{X}, \Theta \neq \mathbb{R}^d$

Averages and minimizers in $\mathbb{R}^d$ have normal limiting laws - great!

Let's turn to non- $\mathbb{R}^d$ settings...

Generalizing averages to $\mathcal{X}, \Theta \neq \mathbb{R}^d$

Averaging a set of observations $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ relies on two essential vector space operations: addition and scalar multiplication.

What would it mean to find the mean of observations when $\mathcal{X}, \Theta \neq \mathbb{R}^d$?

Reframe the mean as a minimizer of a squared distance:

$$\bar{X}_n = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (X_i - \theta)^2 = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n d(X_i, \theta)^2$$

Fréchet means

Definition: (Fréchet, 1948) Let (S, d) be a metric space and P a probability measure on S :

- ▶ *Fréchet function:* $F(p) = \int_S d^2(p, x) dP(x), \quad p \in S.$
- ▶ *Fréchet mean set:* $\mu(P) = \arg \min_{p \in S} F(p)$
- ▶ *Fréchet variance:* $V(P) = \inf_{p \in S} F(p).$

Given $X_1, \dots, X_n \stackrel{iid}{\sim} P$

- ▶ *Sample Fréchet function:* $F_n(p) = \frac{1}{n} \sum_i d^2(p, X_i).$
- ▶ *Sample Fréchet mean set:* $\mu(P_n) = \arg \min_{p \in S} F_n(p)$

For $S = \mathbb{R}^d$ with the L^2 distance, $\mu(P) = \{\mathbb{E}X\}$ and $V(P) = \text{traceVar}(X).$

Existence of Fréchet means

Theorem: Let (S, d) be a proper* metric space and suppose $F(q) < \infty$ for some $q \in S$. Then $\mu(P)$ is nonempty.

Proof: Overview:

- ▶ A continuous function on a compact set attains its infimum.
- ▶ Show $F(p) = \int d^2(p, x) dP(x)$ is Lipschitz (hence, continuous) using Minkowski
- ▶ Show $\{p : F(p) \leq V(p)\}$ is compact.

Note that if $F(q) < \infty$ for some $q \in S$, $F(q) < \infty$ for every q ($F(p) \leq 2F(q) + 2d^2(p, q)$).

* proper = closed bounded sets are compact

Uniqueness fails in general

While we have existence, uniqueness is not guaranteed:

- ▶ $M = S^1$, $d = \text{arc-length}$, and P the uniform distribution.
Then $\mu(P) = S^1$.
- ▶ $M = S^2$, $P = \frac{1}{2}(\delta_N + \delta_S)$ with N, S the North and South Poles. Then $\mu(P)$ is the equator: $\{p : d(p, N) = \pi/2\}$.

Uniqueness requires assumptions on the *support* of P and the *geometry* of M .

Manifolds and intrinsic means

M : complete, connected, C^∞ Riemannian manifold, $\dim M = m$;
 g the metric tensor; d the induced geodesic distance.

- ▶ $\text{Exp}_p : T_p M \rightarrow M$ the Riemannian exponential map;
- ▶ $\text{Log}_p = \text{Exp}_p^{-1}$, defined on $M \setminus C(p)$ where $C(p)$ is the cut locus of p
- ▶ $\text{inj}(M) = \inf_p d(p, C(p))$ the injectivity radius;
- ▶ $\Delta = \sup$ of sectional curvatures

Definition: The *intrinsic mean set* of P on M is $\mu(P)$ for d the geodesic distance.

Definition: Let $j : M \rightarrow \mathbb{R}^D$ be a smooth embedding with $j(M)$ closed. The *extrinsic mean set* is $\mu_j(P) = \arg \min_{p \in S} \int \|j(p) - j(x)\|^2 dP(x)$.

Uniqueness: positive curvature

A local result about uniqueness on manifolds when the curvature $\Delta > 0$:

- If $\Delta > 0$ and $\text{supp}(P) \subseteq \bar{B}(o, r)$ with $r < \frac{1}{2} \min\{\text{inj}(M), \pi/(2\sqrt{\Delta})\}$, F has a unique minimiser in $\bar{B}(o, r)$

Uniqueness: nonpositive curvature

- ▶ If (S, d) is complete, simply connected manifold with $\Delta \leq 0$, then F has a unique minimizer.
- ▶ If S is a complete CAT(0) metric space, then F has a unique minimizer.
 - ▶ Globally nonpositively curved

Next time

Limiting behaviour of Fréchet means on manifolds - meaning and results. . .

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Asymptotic statistics on stratified spaces

Lecture 1: Central limit theory and Fréchet means

Amy Willis

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